

Agenda:

1. LLD Interviews
2. Types of LLD interviews
3. How to approach LLD interview
4. Design a pen.

LLD Interviews

↳ OOPS, SOLID, Design Patterns.

<u>Theoretical</u>	<u>Design</u>	<u>Machine Coding</u>
[Old/Big Traditional Tech MNCs] - Walmart, Oracle, Adobe, Banks - Knowledge based questions - OOPS, SOLID, Design Patterns. - Code very specific sample/example code. - 30-45 mins.	[Modern Big MNCs, MAANG, PayPal, Atlassian] - Design system 'x' ↓ Youtube, Daytm, Uber, Swiggy - Requirement gathering. - class Diagram BMS ↳ design patterns - schema design - APIs design - Code a few classes from class diagram.	[Tech Startups, CRED, Zomato, Zerodha, Coindcx etc.] - Complete requirement doc. - class Diagram ↳ + Design Patterns - schema Design - API design - code all entities - code one feature end-to-end & then move on to other.

- 30-45 min

- 2-5 hrs

Time Distribution for Morning Coding round :

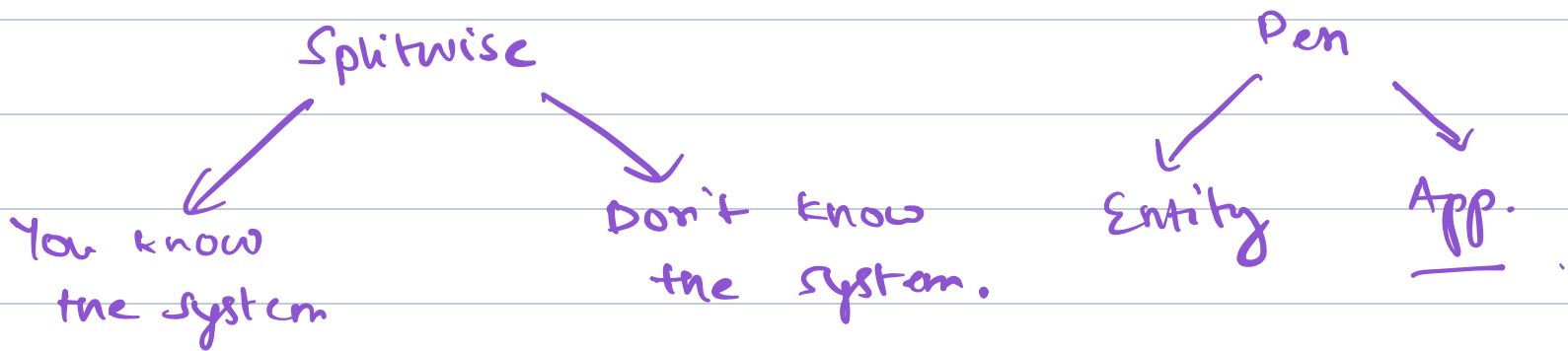
- First 25-30 min :
 - Clarifying the requirements
 - Class Diagram
 - Schema Design
- Next 1.5 hrs
 - Coding .
- Last 30 min - Interviewer to check your code & ask follow up questions .

What we do at scalar:

1. Single line design questions
2. Requirement Gathering .
3. Class diagram + Dfs + Schema design
4. End-to-End important feature coding .

Design a Pen

Step 0: Getting an overview



- 1) What type of system I am building?
- Entity (just entity class design)
 - An application → ask to explain the working of the app.

- 2) Do I need to persist the data
- In-memory / Database.

- 3) How do I take input:
- Hard coded / Command line / REST API.
 - Pattern

Step 1: Gather Requirements

- You are leading the interview - you should suggest ideas with rationale and in the breadth of your knowledge
- Main goal - 5-8 core features. (10 mins)
- login, Register,
- Future requirements...

Visualisation



Requirements of a Pen (Identify all nouns)

- 1) Pen is a physical entity which can write.
- 2) Supporting types - we will support ball pen, gel pen, fountain Pen.
- 3) Every pen will have a name, brand, price, type.

4) Gell pen & Ball pen will have a refill. Fountain Pen will not have a refill.

5) Refills are made of Ink & nib.

6) Nibs will have different radius

7) Ink can have different colors.

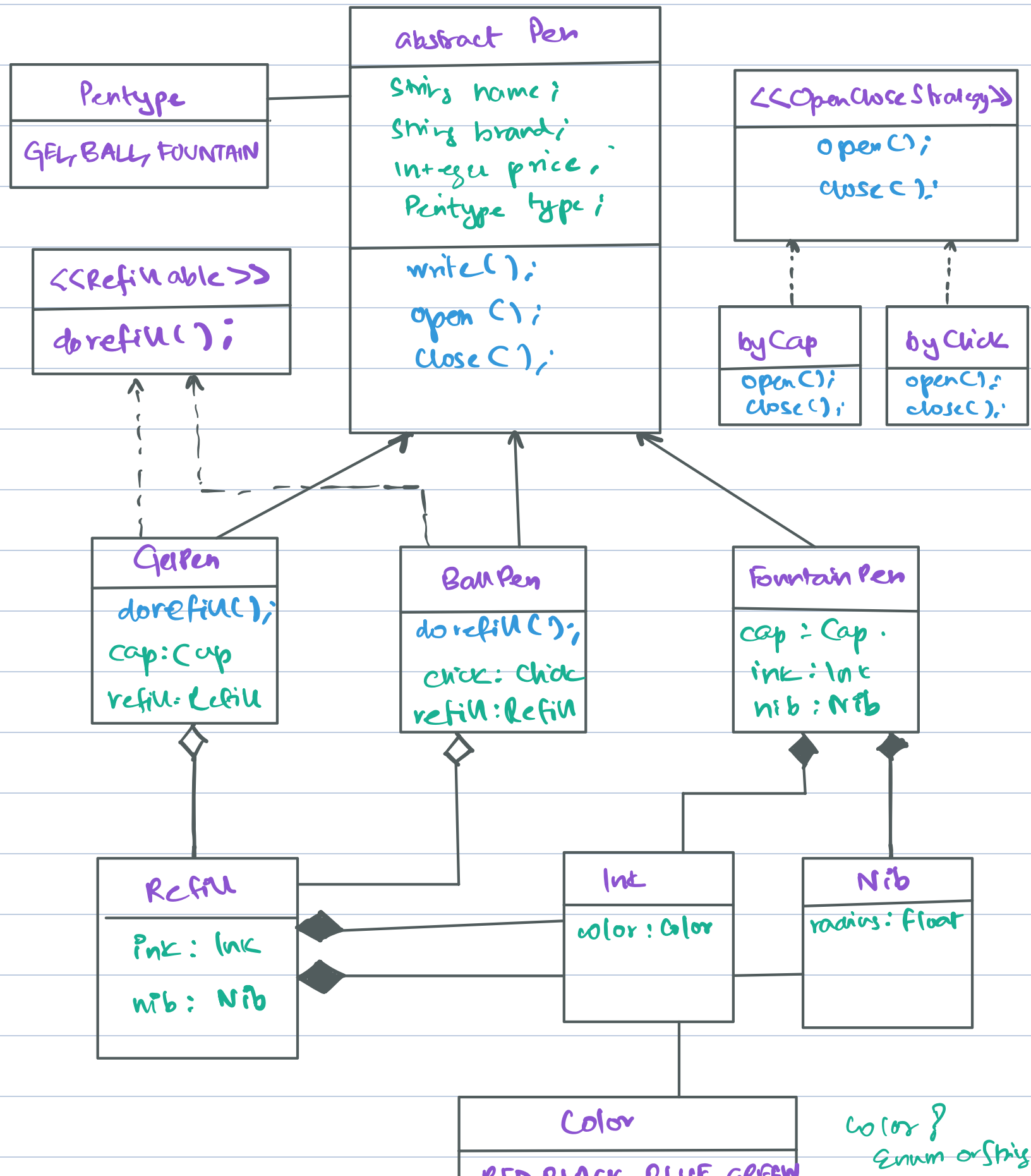
8) Some pens will have cap and some will have click button

9) Pen should be able to write

10) Pen should be able refill

11) Pen should be able close & open

Class Diagram:



RED, BLUE, BLUE, GREEN

limited ↙

↓
unlimited
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